

MIDTERM · BASELINE MODELS

Adilet Search

Finding the article of Kazakhstan law
that answers a question

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Task

Return the article that answers a question in Russian or Kazakh, searching all in-force codes and laws.

Formulation

Known-item retrieval: an article's official title is the query, the article is its one relevant answer. 41,648 labelled queries, no manual annotation.

Research question

How well do classic lexical methods find the right article, per language, and where do they fail?

DATA

AdiletCodex v1.0: 343 acts at article level

65,825

article records in RU, KK and EN (24 codes, 319 laws)

44,216

RU + KK articles used: 22,117 + 22,099

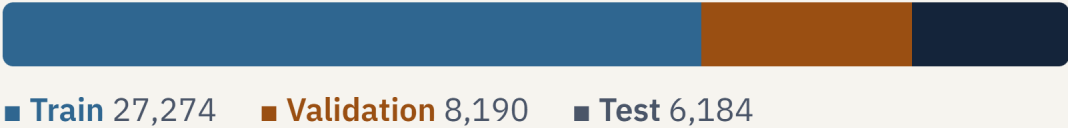
99.3%

of Russian articles have a Kazakh counterpart

27.8%

of articles exceed 512 tokens, so they must be chunked

Split by act, no leakage



Documents 70/15/15 grouped by doc_id, stratified by doc_type, seed 42.
All language versions of an act stay in one subset.

Cleaning; statute text stays verbatim

Issue fixed in search_text	Rows
Amendment footnotes moved out of the norm	8.5%
Repeated whitespace collapsed	24%
Latin i/I inside Kazakh words replaced	9.4%

BASELINE

Three lexical baselines, from unsupervised to learned

1 · Unsupervised

BM25

IDF fitted on the corpus; k_1 and b tuned on validation with a 5×4 grid. Snowball stemming for Russian, plain tokens for Kazakh.

2 · Unsupervised

TF-IDF cosine

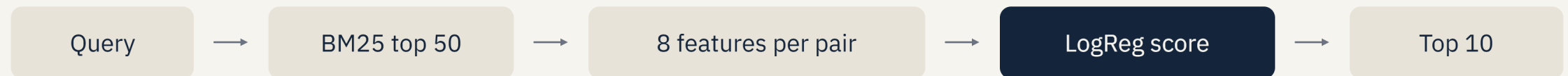
Word 1–2-grams, sublinear tf. Shows what BM25's term saturation and length normalisation add.

3 · Learned

LogReg reranker

Logistic Regression on 8 lexical features reorders BM25's top 50. Balanced classes, C chosen on validation, 4,000 train queries per language.

TWO-STAGE PIPELINE OF THE LEARNED BASELINE



METRICS

Ranking metrics, because users read the top of the list

nDCG@10 • **primary**

Rewards the right article near the top, with a position discount; extends to graded human labels later.

Recall@10

Share of queries whose answer is on the first page of results.

Recall@50

The reranker's ceiling: it can only reorder BM25's top 50.

MRR@10

How soon the first correct article appears in the list.

Reranker as a classifier

ROC-AUC, PR-AUC and F1 on (query, article) pairs.

Why not accuracy

About 22,000 wrong articles per query: rejecting everything would score above 99.9%.

RESULTS · TEST SET

The learned reranker is best in both languages

nDCG@10 ■ Russian ■ Kazakh

BM25, default (k1 1.2, b 0.75)



BM25, tuned on validation



TF-IDF cosine



TF-IDF + LogReg reranker (best)



Bars drawn on a 0–0.7 scale from zero.

Recall

System	@10 RU	@10 KK	@50 RU	@50 KK
BM25 default	0.727	0.689	0.841	0.807
BM25 tuned	0.736	0.693	0.846	0.815
TF-IDF cosine	0.698	0.663	0.846	0.812
LogReg reranker	0.751	0.726	0.846	0.815

Reranker as a classifier

ROC-AUC 0.930 · PR-AUC 0.382 · F1 0.204 on 309,200 test pairs, only 1.66% positive. It orders candidates well; only the order is used.

CONCLUSIONS

Lexical search works only when the question uses the statute's words

75% / 73%

of RU / KK queries get the right article in the top 10

26.2%

of test queries miss the top 10 (1,620 of 6,184)

15–19%

of answers are not even in BM25's top 50

Generic titles. Titles shared by 3+ acts fail 83–84% of the time: 11% of queries, 35% of failures.

Word overlap. Failed queries share fewer words with their article: median 0.67 vs 1.00.

Kazakh is harder. No stemmer: 59,296 vs 18,311 terms. Character n-grams narrow the gap.

NEXT STAGE

Close the vocabulary gap with semantic retrieval

1

Dense + hybrid

Multilingual E5 embeddings over ~400-token chunks, fused with BM25 by RRF. Same split, same metrics.

2

Cross-encoder

A reranker that reads query and article together, over the hybrid top 50. Targets competing neighbouring articles.

3

Human gold set

Plain-language questions with graded labels, RU and KK scored separately: real product quality.

ALREADY RUNNING: END-TO-END PROTOTYPE

FastAPI backend + Qdrant + React UI (RU / KK / EN). Demo index: 9,259 real articles from 16 acts with multilingual-e5-base; answers stream with [1]-style citations. Not yet evaluated on the gold set.